

Professional Abridgement

Reliable and resourceful Design Verification Engineer with a strong record of corporate client satisfaction. Adept multi-tasker who thrives in a high-pressure work environment. Able to explain complicated concepts clearly to a variety of professional and non-professional audiences.

Experience Summary

- VLSI Verification Engineer with 8 years of experience in IP Verification and SoC verification.
- Good programming knowledge of Verilog, System Verilog, VHDL, C/C++.
- Proficient in scripting languages: Make, Perl, Python.
- Debug, report, and work closely with design engineers.
- Creating and maintaining regression test suites.
- Have knowledge on Assertions.
- Create test case to ensure maximum coverage.
- Develop advanced verification environment and test bench components in System Verilog using VMM or UVM methodology.
- Have good knowledge on protocols like UART, SPI, I2C.

Work Experience

- 5 years of Senior Design verification work experience Engineer with HCL Technologies., Bengaluru as a Senior verification Engineer Since 11 Nov 2019.
- 3 Years of Design verification Engineer with ANANTH TECHNOLOGIES Ltd., Hyderabad from 26 Sep 2016 to 9 Nov 2019.

Educational Qualifications:

Qualification	Institution/Board	Year of Passing	Percentage
M.Tech	SRM University	2016	77.08
B.Tech	Holy Mary Institute of Science and Technology	2013	70.80
Intermediate	Vivekananda Junior college for girls	2009	70.90
Secondary School of Education	Vikram Public School	2007	65

EDA Tools and Expertise:

- TB Methodology : **UVM**
- Hardware Verification language : **System Verilog**
- Hardware Description language : **Verilog, VHDL**
- Simulation tools : **QuestaSim, Cadence Incisive, Synopsys VCS, Modelsim ,Verdi**
- Bus protocols : **SPI, UART, AHB, AXI, APB protocols, I2C.**
- Operating System : **Linux, Windows.**
- Scripting languages : **Python, Perl, TCL**

Project Summary:**1. RZV2H_(Renesas_ Japan):**

Role : Verification Engineer
HVL : SV ,C++
Tools : Synopsys VCS, Verdi

Description:

The objective of this project is to verify the functionality of ICU for RZV2H. ICU generates the event described in event list “Event_List file” and interrupt, the event output, and the error reset trigger are generated masking, logical reversing, and maintaining the factor by using a built-in register.

Responsibilities:

- Executed the verification test plan by developing checkers, stimulus and coverage using System Verilog / C++ and running simulations
- Developed assertion based test cases for ICU.
- Developed Python scripts.
- Debug, report, and work closely with design engineers.
- Communicate with the team and execute the test plans in timely matter.
- Creating and maintaining regression test suites

2. System Veification Development Phase14(Renesas_Japan):

Role : Verification Engineer
Tools : Helium

Description:

This project Project verifies the model present in the Helium model catalog.Platforms can run either from the Helium Platform Assembly GUI or usig the command line. In Helium Platform Assembly GUI, all the models and platforms are available in the Catalog Browser tab in Helium Platform Assembly.

Responsibilities:

- Developed Test plan and test case implementation.
- Ran the platform present in Helium model catalog.
- Prepared the reports and procedure documents.

3. System Veification Development Phase14(Renesas_Japan):

Role : Verification Engineer
Tools : Joules,Palladium

Description:

This project performs Dynamic power analysis and power estimation for RZV2N using JOULES.

Responsibilities:

- Developed Test plan and test case implementation.
- Ran the test cases for RZV2N using joules and analyzed the generated reports and waveforms.
- Prepared the reports and procedure documents.

4. HS-PRO Control Panel_(HONEYWELL_ Bengaluru):

Role : Verification Engineer
HDL/HVL – Methodology : SV ,UVM,
Tools : Mentor graphics -Questa

Description:

This project manages detection, Notification and extinguishing functions. It contains the ADC other peripheral devices like RGB LED driver, HART Modem are connected to FPGA through SPI Interface. FPGA communicates with the IO controller through CAN Controller. FPGA shall interface with CAN IP through generic protocol APB.

Responsibilities:

- Developed Test plan and Verification Specification Creation.
- Developed the test cases and the test bench Components from Scratch and.
- Block Level and Integration Level Simulation, debugging.
- Developed Python scripts.

5. Radar Reciever_(DARE_ Bengaluru).

Role : Verification Engineer
HDL/HVL – Methodology : SV ,UVM
Tools : Mentor graphics -Questa, MATLAB

Description:

The objective of this project is to verify the functionality of FPGAs with ADC interface (JESD Rx and Tx Configuration) in Wide Band and Narrow band. PCIe (gen3) Interface is used for communication.

Responsibilities:

- Requirement capture and test case document preparation.
- Developed the Verification Plan, test bench Components from Scratch.
- Simulation, debugging and register configuration through PCIe Interface.

6. Falcon 6X Verification_(MEGGITT_akron):

Role	:	Verification Engineer
HDL/HVL – Methodology	:	VHDL
Tools	:	Mentor graphics Questa

Description:

The objective of this project is to verify the functionality of FPGAs in Braking control and Landing Gear Steering Control Mechanism

Responsibilities:

- Developed the Verification Plan, test bench Environment and test suit.
- Created hardware verification procedure by identifying the test scenarios for the requirements.
- Simulation and debugging of complete verification.
- Developed Perl/Bash scripts.

7. Verification UART controller for SoC**Description:**

Connectivity level verification of UART is performed on SoC which is connected to a multi processor through an AXI bus interface. Programmed bit file on ZCU104 board and developed C testcases. Connectivity is verified between processors and UART.

Tools : Xlinix Vivado SDK

Language: C++

Responsibilities:

- Developed C++ testcases at Top/Sub-system/Block-levels.
- Developed Python scripts.
- Develop verification IP which can be reused at different levels of verification: block level, sub-system level, SoC level.

Harsha

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8. Digital Quad Channel Controller Design and Verification_(RCI, Hyderabad)

Role	:	Verification Engineer
HDL/HVL – Methodology	:	SV, UVM
Tools	:	Mentor graphics Questa

Description:

This project main concern is to develop the Digital Quad channel controller unit with ADC and DAC interface for FPGA, with configuration UART data. Arbitrary inputs are given and output observed.

Responsibilities:

- Developed UVM based testbenches at Top/Block-levels.
- Developed test plan and test cases focused on thoroughly testing the UART Controller working in a UVM environment.
- Developed verification IP which can be reused at different levels of verification: block level, sub-system level, SoC level, etc.
- Developed functional coverage model
- Implemented UVM test bench for random testing.
- Developed scoreboards and protocol checkers to check packet data
- Worked on functional coverage analysis